Practical Name: Automated Login Testing

Aim: To create automated login functionality, authentication validation, assertions and exception handling scenarios using Selenium WebDriver with Python.

Source Code:

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
from selenium.common.exceptions import (
    NoSuchElementException,
    TimeoutException,
    ElementNotInteractableException,
    StaleElementReferenceException
)

URL = "https://www.saucedemo.com/"

VALID_USERNAME = "standard_user"
VALID_PASSWORD = "secret_sauce"


def start_browser():
    driver = webdriver.Chrome()
    driver.maximize_window()
    return driver


def login(driver, username, password):
    driver.get(URL)
    wait = WebDriverWait(driver, 10)

    username_box = wait.until(
        EC.visibility_of_element_located(
            (By.ID, "user-name")
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        )
    )

    password_box = wait.until(
        EC.visibility_of_element_located(
            (By.ID, "password")
        )
    )

    login_button = wait.until(
        EC.element_to_be_clickable(
            (By.ID, "login-button")
        )
    )

    username_box.clear()
    username_box.send_keys(username)

    password_box.clear()
    password_box.send_keys(password)

    login_button.click()


# TC-01 Valid Login
def test_valid_login():
    driver = start_browser()

    try:
        login(driver, VALID_USERNAME, VALID_PASSWORD)

        WebDriverWait(driver, 10).until(
            EC.url_contains("inventory")
        )

        assert "inventory" in driver.current_url

        products_title = WebDriverWait(
            driver, 10
        ).until(
```

```
            EC.visibility_of_element_located(
                (By.CLASS_NAME, "title")
            )
        )

        assert products_title.text == "Products"

        print("TC-01 Valid Login: PASS")
        print("Authentication Assertion: PASS")

    except AssertionError:
        print("TC-01 Valid Login: FAIL")

    except NoSuchElementException:
        print("NoSuchElementException handled")

    except TimeoutException:
        print("TimeoutException handled")

    except ElementNotInteractableException:
        print("ElementNotInteractableException handled")

    except StaleElementReferenceException:
        print("StaleElementReferenceException handled")

    finally:
        driver.quit()


# TC-02 Invalid Password
def test_invalid_password():
    driver = start_browser()

    try:
        login(driver, VALID_USERNAME, "wrong_password")

        error_message = WebDriverWait(
            driver, 10
        ).until(
            EC.visibility_of_element_located(
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```
            (By.CSS_SELECTOR, "h3[data-test='error']")
        )
    )

    assert (
        "Username and password do not match"
        in error_message.text
    )

    print("TC-02 Invalid Password: PASS")

except AssertionError:
    print("TC-02 Invalid Password: FAIL")

except TimeoutException:
    print("Error message was not displayed")

finally:
    driver.quit()


# TC-03 Invalid Username
def test_invalid_username():
    driver = start_browser()

    try:
        login(driver, "wrong_user", VALID_PASSWORD)

        error_message = WebDriverWait(
            driver, 10
        ).until(
            EC.visibility_of_element_located(
                (By.CSS_SELECTOR, "h3[data-test='error']")
            )
        )

        assert (
            "Username and password do not match"
            in error_message.text
        )
```

```
        print("TC-03 Invalid Username: PASS")

    except AssertionError:
        print("TC-03 Invalid Username: FAIL")

    except TimeoutException:
        print("Error message was not displayed")

    finally:
        driver.quit()


# TC-04 Empty Credentials
def test_empty_credentials():
    driver = start_browser()

    try:
        driver.get(URL)

        login_button = WebDriverWait(
            driver, 10
        ).until(
            EC.element_to_be_clickable(
                (By.ID, "login-button")
            )
        )

        login_button.click()

        error_message = WebDriverWait(
            driver, 10
        ).until(
            EC.visibility_of_element_located(
                (By.CSS_SELECTOR, "h3[data-test='error']")
            )
        )

        assert "Username is required" in error_message.text
```

```
        print("TC-04 Empty Credentials: PASS")

    except AssertionError:
        print("TC-04 Empty Credentials: FAIL")

    except TimeoutException:
        print("Validation message was not displayed")

    finally:
        driver.quit()


# TC-05 Logout
def test_logout():
    driver = start_browser()

    try:
        login(driver, VALID_USERNAME, VALID_PASSWORD)

        WebDriverWait(driver, 10).until(
            EC.url_contains("inventory")
        )

        menu_button = WebDriverWait(
            driver, 10
        ).until(
            EC.element_to_be_clickable(
                (By.ID, "react-burger-menu-btn")
            )
        )

        menu_button.click()

        logout_button = WebDriverWait(
            driver, 10
        ).until(
            EC.element_to_be_clickable(
                (By.ID, "logout_sidebar_link")
            )
        )
```

```
        logout_button.click()

        WebDriverWait(driver, 10).until(
            EC.visibility_of_element_located(
                (By.ID, "login-button")
            )
        )

        assert "saucedemo.com" in driver.current_url

        print("TC-05 Logout: PASS")

    except AssertionError:
        print("TC-05 Logout: FAIL")

    except TimeoutException:
        print("Logout operation timed out")

    finally:
        driver.quit()


# Execute all test cases
test_valid_login()
test_invalid_password()
test_invalid_username()
test_empty_credentials()
test_logout()
```